

FINS5545 FINTECH PROJECT B

FinMosaic

Funds, sentiment analytics and a personalised financial-information app

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1. The funds and the backtest design

FinMosaic contains ten investable backtest histories: four combined equity-crypto funds, three equity-only funds and three crypto-only funds. The combined panel contains 50 equities and 10 cryptocurrencies. Equal Weight is the transparent benchmark; the optimised methods are long-only Minimum Variance and Maximum Sharpe. A fourth combined fund applies the fixed VADER sector tilt described in Section 4.

FinMosaic is aimed at self-directed investors who want one place to compare the market views they have deliberately assembled. Rather than imposing a single generic screen, it lets a user combine equities, crypto assets, sector context and systematic fund histories. The three universes answer different comparison questions: equity-only isolates listed-company risk, crypto-only exposes the native seven-day market, and the combined universe shows the effect of holding both. The product is an information and comparison tool, not a suitability assessment or investment recommendation.

Walk-forward design

Equity-only and combined funds use a trailing 252-equity-trading-day estimation window, the equity trading calendar and 252-day annualisation. Their live sample contains 752 returns from 5 January 2021 to 29 December 2023. Crypto-only funds use a trailing 365-calendar-day window, retain weekend dates and use 365-day annualisation; their live sample contains 1,094 returns from 2 January 2021 to 31 December 2023.

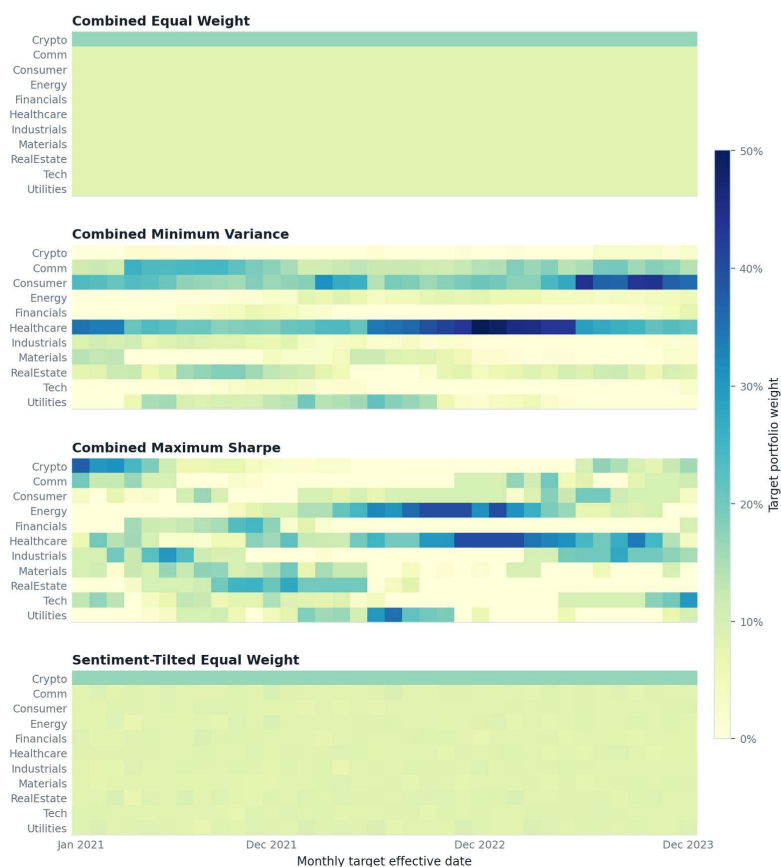
Rebalancing occurs monthly on the first observed date. The estimation sample ends before the rebalance date, and each target becomes effective on the next observed date. Returns accrue only after formation; holdings then drift until the next rebalance. All funds are long-only and fully invested. Transaction costs and the risk-free rate are fixed at zero.

For return vector μ and covariance matrix Σ , Minimum Variance solves $\min(w' \Sigma w)$. Maximum Sharpe solves $\max((w' \mu) / \sqrt{w' \Sigma w})$ because the risk-free rate is zero. Both impose $\sum(w) = 1$ and $w \geq 0$. Maximum Sharpe additionally imposes $w_i \leq 0.10$. The window, frequency and cap were fixed before evaluating out-of-sample performance. (Markowitz, 1952; Sharpe, 1966).

FINMOSAIC / MONTHLY TARGET WEIGHTS

Dynamic methods departed from Equal Weight in different ways

Ten equity sectors plus unchanged or optimized crypto sleeve; first effective date each month



Source: results/data/fund_weights.csv; author calculations. Sample: monthly targets effective 5 Jan 2021-4 Dec 2023. Units: percent of portfolio.

Figure 1. Monthly target weights over time for the dynamic combined Minimum Variance, Maximum Sharpe and Sentiment-Tilted funds relative to Combined Equal Weight, 5 Jan 2021-29 Dec 2023. Weights are portfolio percentages. Source: results/data/fund_weights.csv; author calculations.

Figure 1 confirms that the dynamic methods produce different targets across rebalances. At the final rebalance, combined equity/crypto sleeve weights are 83.33%/16.67% for Equal Weight, 84.33%/15.67% for Maximum Sharpe and 97.53%/2.47% for Minimum Variance.

The design makes the timing rule easy to audit, but it also makes deliberate simplifications. Equal Weight is transparent and diversified, whereas optimisation can react to estimated means and covariances that are unstable in a finite historical window. The 10% Maximum Sharpe cap limits single-asset concentration, while the long-only constraint rules out short positions. These choices make the alternatives comparable, but they do not establish that one rule will remain superior in a future market.

2. Out-of-sample results and fund fact sheets

Table 1. Walk-forward out-of-sample performance for all funds. Returns, volatility and maximum drawdown are percentages; growth is the terminal US-dollar value of an initial \$1. Equity/combined sample: 5 Jan 2021-29 Dec 2023; crypto sample: 2 Jan 2021-31 Dec 2023. Source: results/tables/fund_comparison_report_table.csv.

Universe	Fund / method	Return	Vol.	Sharpe	Max DD	\$1 end
Combined	Equal Weight	15.47%	21.46%	0.778	-27.74%	\$1.536
Combined	Minimum Variance	5.43%	12.77%	0.478	-15.07%	\$1.171
Combined	Maximum Sharpe	17.33%	21.83%	0.841	-19.44%	\$1.611
Combined	Sentiment Tilt	15.21%	21.43%	0.768	-27.57%	\$1.526
Equity-only	Equal Weight	13.29%	16.11%	0.855	-20.19%	\$1.451
Equity-only	Minimum Variance	5.36%	12.74%	0.473	-14.95%	\$1.168
Equity-only	Maximum Sharpe	10.81%	16.64%	0.700	-19.23%	\$1.359
Crypto-only	Equal Weight	37.11%	81.74%	0.799	-81.72%	\$2.576
Crypto-only	Minimum Variance	28.54%	64.71%	0.713	-76.04%	\$2.122
Crypto-only	Maximum Sharpe	37.11%	81.74%	0.799	-81.72%	\$2.576

Table 1 reports all ten funds. Combined Maximum Sharpe records the highest combined-fund annualised return (17.33%), Sharpe ratio (0.841) and terminal value (\$1.611). Combined Minimum Variance records the lowest combined volatility (12.77%) and shallowest combined maximum drawdown (-15.07%), with a 5.43% annualised return.

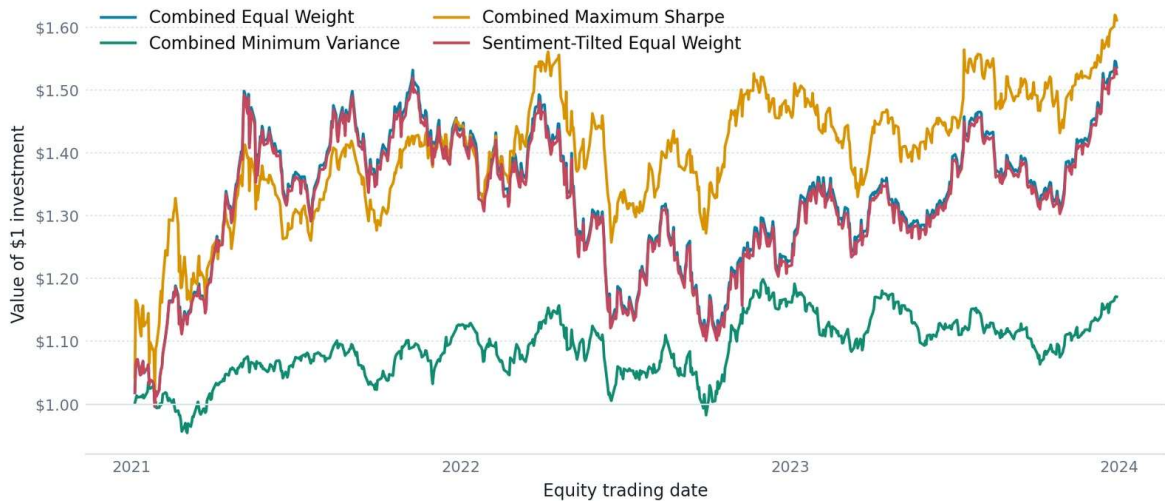
Equity Equal Weight has the highest Sharpe ratio across all funds (0.855). Equity Maximum Sharpe records 10.81% annualised return and a 0.700 Sharpe ratio, below Equity Equal Weight's 13.29% and 0.855. Crypto Equal Weight and capped Crypto Maximum Sharpe are identical: both return 37.11% annualised, grow \$1 to \$2.576 and experience an -81.72% maximum drawdown. Crypto Minimum Variance lowers volatility to 64.71% and drawdown to -76.04%, while its annualised return is 28.54%.

The results show a clear return-risk trade-off rather than a universal winner. Minimum Variance consistently reduces volatility and the size of the deepest loss, but it gives up much of the higher-return exposure. Maximum Sharpe uses estimated return as well as covariance information, so its targets can differ from Equal Weight; over this sample it improves the combined result but not the equity-only benchmark. Crypto dominates the largest terminal-growth outcomes while also carrying drawdowns that are materially larger than those of the equity and combined funds.

FINMOSAIC / FUND COMPARISON

Four funds produced distinct out-of-sample paths

Growth of \$1 from simple daily portfolio returns; no transaction costs



Source: Hosted course price and headline data; author calculations. Out-of-sample period: 5 Jan 2021-29 Dec 2023. Units: US dollars

Figure 2. Growth of an initial \$1 for the four combined funds, 5 Jan 2021-29 Dec 2023. Values are US dollars and transaction costs are zero. Source: results/data/fund_returns.csv; author calculations.

Figure 2 shows the path behind the terminal values in Table 1. Combined Maximum Sharpe finishes at \$1.611, Combined Equal Weight at \$1.536, Sentiment-Tilted Equal Weight at \$1.526 and Combined Minimum Variance at \$1.171. Appendix Figure A11 compares drawdowns, and Appendix Figure A12 compares Sharpe ratios for the same four combined funds.

Appendix Figures A1-A10 provide one report-ready fact sheet per fund, including the saved method assumptions, growth, maximum drawdown and latest target holdings. The fact sheets confirm that the latest Crypto Minimum Variance target is concentrated in TRX (50.13%), BTC (47.78%) and ETH (2.09%).

3. The sentiment index

Standard NLTK VADER scores each original headline before aggregation. The build averages headline compound scores to ticker-day and then equal-weights observed ticker-days within each sector. It does not strip text or add finance-specific terms. No-news ticker-days and sector-days remain missing with zero coverage; an observed score of exactly zero remains a genuine neutral observation. (Hutto & Gilbert, 2014).

Only the one-equity-trading-day-lagged sector signal is eligible for later portfolio use. A weekend or Monday headline aligned to Monday is therefore first usable on Tuesday. The sector index covers 10,060 sector-days: 9,832 have observed sentiment and 228 are missing. Technology is observed on all 1,006 equity trading dates; Materials and Real Estate each have 940 observed dates and 66 no-news dates.

VADER scores 146,830 headlines. The saved thresholds classify 19,064 (12.98%) as negative, 72,786 (49.57%) as neutral and 54,980 (37.44%) as positive. The complete company grid has 50,300 ticker-days, of which 37,962 (75.47%) contain news; 8,724 observed ticker-days have a genuine score of exactly zero.

Using the original supplied headline text preserves a traceable link between each score and its source input. The equal-weight ticker step means that a company with many headlines does not automatically dominate its sector signal. Treating no-news as missing is equally important: replacing missing observations with zero would make unavailable information look neutral and would hide differences in coverage. These choices make the index interpretable as observed news tone, rather than as a complete description of every firm or sector.

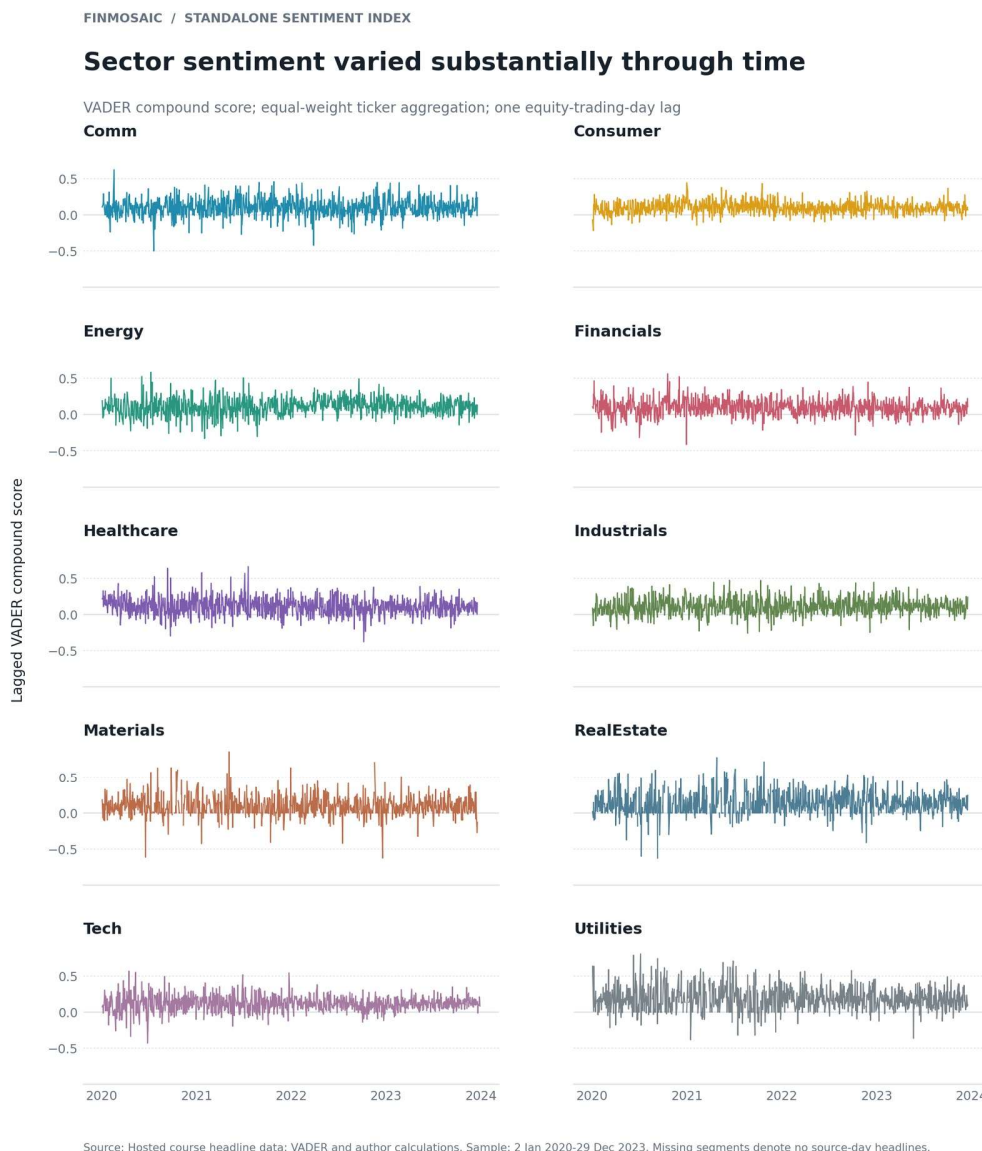


Figure 3. Daily VADER compound sentiment by equity sector, 2 Jan 2020-29 Dec 2023. Scores range from -1 to +1 and sector-days without headlines remain missing. Source: results/data/sector_sentiment_index.csv; author calculations.

Figure 3 displays the standalone sector series rather than a trading recommendation. Appendix Figure A13 reports sector news coverage, while Appendix Figure A14 gives a more readable monthly company-level view for a cross-sector watchlist. These company results remain a drill-down and are not used as a proxy for an entire fund.

The Loughran-McDonald 2018 dictionary is retained as a separate robustness check. It uses the saved 2019-03 version and the score (positive tokens - negative tokens)/alphabetic tokens, with case-insensitive unigrams and no phrase scoring. The formal paper and archived resource metadata require personal verification before citation. (Loughran & McDonald, 2011).

4. The fusion extension and whether it adds value

The fusion fund starts from Combined Equal Weight. Within the equity sleeve, sector s has raw weight $w_s \times (1 + 0.50 \times S_{s,t-1})$, where $S_{s,t-1}$ is the saved one-trading-day-lagged VADER sector signal. The equity weights are renormalised to preserve the base equity sleeve, and every crypto target is copied unchanged. A missing signal receives multiplier one but remains missing in the stored sentiment data. The strength 0.50 was fixed ex ante and was not tuned on out-of-sample performance.

Table 2. Combined Equal Weight before and after the fixed sentiment tilt, 5 Jan 2021-29 Dec 2023. Returns, volatility and drawdown are decimal rates; growth is the terminal value of \$1. Source: results/tables/fusion_comparison.csv.

Metric	Base	Tilted	Tilt minus base
Annualised return	15.47%	15.21%	-0.259%
Annualised volatility	21.46%	21.43%	-0.030%
Sharpe ratio	0.778	0.768	-0.010
Maximum drawdown	-27.74%	-27.57%	+0.173%
Growth of \$1	\$1.536	\$1.526	-\$0.010

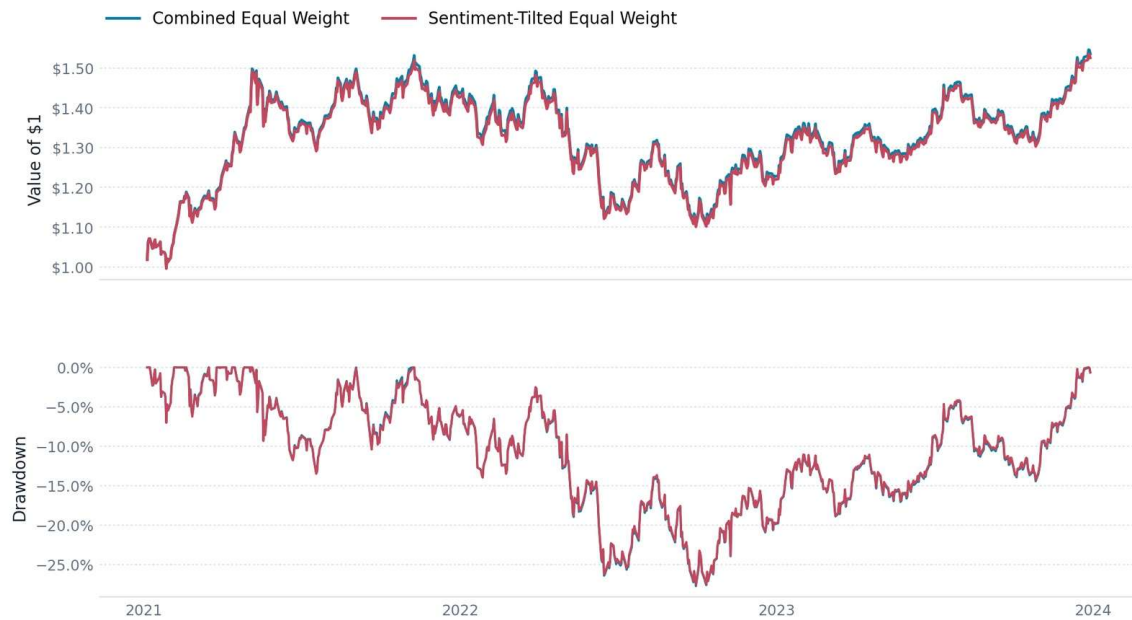
Table 2 shows a negative return result. Relative to Combined Equal Weight, the tilt lowers annualised return by 0.259 percentage points, lowers Sharpe by 0.0097 and reduces terminal growth by \$0.0103 per initial \$1. Annualised volatility falls by 0.030 percentage points and maximum drawdown improves by 0.173 percentage points. On return, Sharpe and terminal growth, the fixed rule does not add out-of-sample value over this sample.

The small gap is consistent with the construction of the rule. The tilt only reallocates the existing equity sleeve across sectors, preserves its total weight, leaves crypto unchanged and uses a single lagged score per sector. Monthly rebalancing and diversified sector aggregation also smooth individual headline moves. The result is useful precisely because it is negative: under this fixed, untuned specification, the news signal did not produce a material improvement in the main out-of-sample return measures.

FINMOSAIC / SENTIMENT FUSION

Fixed sentiment tilt versus Combined Equal Weight

Monthly targets; 50% tilt strength; crypto targets unchanged; zero transaction costs



Source: Hosted course price and headline data; author calculations. Out-of-sample period: 5 Jan 2021-29 Dec 2023.

Figure 4. Combined Equal Weight before and after the fixed lagged-sector sentiment tilt, 5 Jan 2021-29 Dec 2023. Panels show growth of \$1 (US dollars) and drawdown (percent). Source: results/data/fund_returns.csv and results/tables/fusion_comparison.csv; author calculations.

Figure 4 shows that the two paths remain close throughout the common sample. Appendix Figures A15-A17 compare VADER with Loughran-McDonald. LM classifies 75.43% of headlines as neutral versus 49.57% for VADER and has lower score dispersion (0.0627 versus 0.2881).

Only 38,347 of 146,830 headlines contain an LM match (26.12%), and 45,354 of 1,374,846 tokens match (3.30%). Mean sector Pearson correlation is 0.512 and mean classification disagreement is 46.35%. This dictionary output is a read-only methodology comparison; it does not replace VADER or enter a second fused portfolio.

The robustness comparison reduces confidence in any strong interpretation of a single sentiment score. VADER and Loughran-McDonald have different vocabularies, score distributions and coverage, and their classifications frequently disagree. This does not prove either method is wrong; it shows that dictionary choice is a material modelling assumption. For that reason, the app presents the comparison as methodology evidence, while the fused portfolio remains based on one pre-specified VADER rule.

5. The app and the investor journey

FinMosaic is a results-only Streamlit application. Its validated loader reads precomputed CSV and figure artifacts under results/; it does not load raw prices or execute returns, optimisation, VADER, Loughran-McDonald or fusion logic. The cached bundle is versioned so a schema change invalidates stale app data.

My Mosaic is the default workspace and carries the selected fund, equities, crypto assets and sectors across views. It combines saved asset growth and risk, sector VADER coverage, fund metrics and latest holdings. Funds supports universe filtering, historical growth, drawdown, return-risk comparison, assumptions, latest holdings, hypothetical allocation exposure and a precomputed portfolio-news context series.

Watchlist uses saved native-calendar asset histories. News defaults to monthly observed-only company and sector VADER means, retains missing no-news months, shows headline volume and offers short 30- or 90-day daily views. Methodology presents the VADER/LM comparison as read-only robustness evidence. Every workspace states that outputs are historical and are not investment advice.

The target user is a self-directed investor who wants to organise historical financial information around assets and sectors they already follow, without receiving a buy or sell recommendation. A typical journey begins by choosing a universe and comparing fund fact sheets. The user can then inspect the selected fund's growth, drawdown and latest holdings, create a watchlist of equities and crypto assets, and view company or sector news context with explicit coverage information. The final allocation workspace reports hypothetical historical exposure rather than generating advice.

The interface uses a shared ?mosaic? of selections so that a chosen fund, watchlist and related sector context remain connected across views. This reduces repeated filtering and makes it clearer why a news or holdings panel is being shown. Default monthly sentiment views emphasise readable variation and coverage; short daily windows remain available for closer inspection. The design deliberately separates missing news from neutral tone and repeatedly labels the outputs as historical information only.

6. Critical reflection with three concrete recommendations

The project establishes a reproducible comparison framework rather than a predictive investment product. The walk-forward timing, native-calendar treatment and saved fact sheets make the fund results inspectable. Table 1 and Figure 2 show that the methods differ in economically meaningful ways, while Figure 1 makes the changing targets visible. However, the backtests omit implementation frictions and cover one finite evaluation period. Figures 3 and 4 also show that news-derived information needs careful interpretation: coverage is high at sector level, but dictionary choice and the small fusion result limit the strength of any performance claim.

Recommendation 1 - implementation frictions

A next build should save pre-trade and post-trade weights at every rebalance, calculate one-way turnover, and evaluate fixed bid-ask and fee scenarios before comparing net returns. The cost grid should be fixed before the next out-of-sample period is reviewed and should be reported alongside gross results. This would test whether the apparent differences between target rules survive a more realistic implementation assumption.

Recommendation 2 - portfolio estimation and concentration

A next build should pre-specify a covariance-regularisation method and concentration sensitivity tests, such as tighter per-asset caps or sector caps. It should compare those alternatives in a fresh holdout period rather than selecting the setting that looks best in 2021-2023. This is particularly important for the crypto universe, where the latest Minimum Variance target is heavily concentrated in TRX and BTC.

Recommendation 3 - financial-language validation

A next build should validate sentiment against a manually labelled, finance-relevant headline sample before changing the production signal. It should pre-specify the alternative lexicon or model, document token coverage and evaluate whether the signal improves a future holdout period after costs. The VADER/LM disagreement demonstrates that a more financial vocabulary is plausible, but it is not evidence that an untested replacement would improve the strategy.

FinMosaic therefore provides a reproducible historical foundation for comparing systematic funds, asset watchlists and news context. It establishes what these fixed rules produced in the stated sample, not what an investor should hold or what markets will do next. The remaining uncertainty around costs, estimation error, sentiment measurement and future regimes is part of the product's evidence, rather than something to be concealed.

References

The following method sources were checked against their publisher or proceedings pages on 13 August 2026. Course data are cited in tables and figure captions as hosted course data and author calculations.

Hutto, C. J., & Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216-225. <https://doi.org/10.1609/icwsm.v8i1.14550>

Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *The Journal of Finance*, 66(1), 35-65. <https://doi.org/10.1111/j.1540-6261.2010.01625.x>

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77-91. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>

Sharpe, W. F. (1966). Mutual fund performance. *The Journal of Business*, 39(S1), 119-138. <https://doi.org/10.1086/294846>

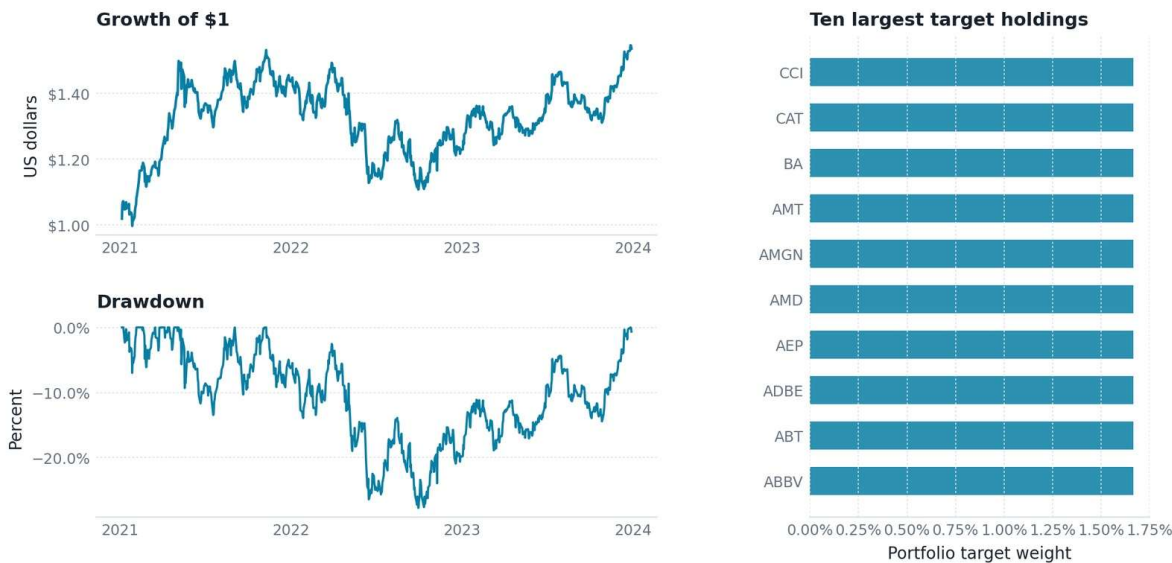
Appendix: Supporting exhibits

FINMOSAIC / FUND FACT SHEET

Combined Equal Weight

Out-of-sample performance: 5 Jan 2021-29 Dec 2023 | Latest target holdings: 4 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
15.5%	21.5%	0.778	-27.7%	\$1.536



Method: Equal Weight | Estimation: 252 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

Appendix Figure A1. Combined Equal Weight fact sheet, 5 Jan 2021-29 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Combined Minimum Variance

Out-of-sample performance: 5 Jan 2021-29 Dec 2023 | Latest target holdings: 4 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
5.4%	12.8%	0.478	-15.1%	\$1.171



Method: Min Variance | Estimation: 252 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

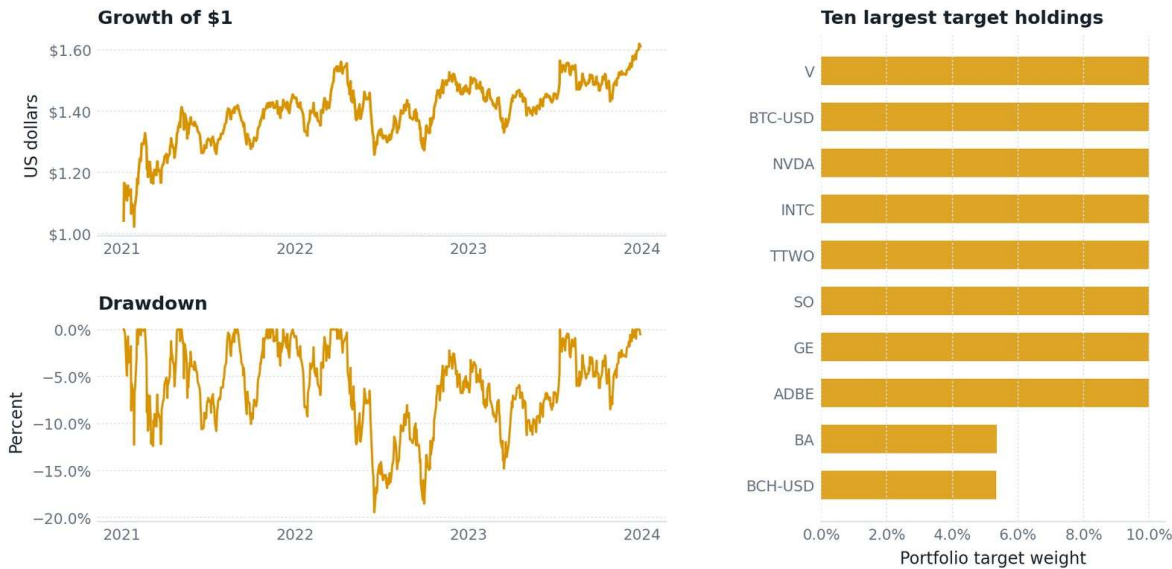
Appendix Figure A2. Combined Minimum Variance fact sheet, 5 Jan 2021-29 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Combined Maximum Sharpe

Out-of-sample performance: 5 Jan 2021-29 Dec 2023 | Latest target holdings: 4 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
17.3%	21.8%	0.841	-19.4%	\$1.611



Method: Max Sharpe | Estimation: 252 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested; maximum 10% per asset | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

Appendix Figure A3. Combined Maximum Sharpe fact sheet, 5 Jan 2021-29 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Sentiment-Tilted Equal Weight

Out-of-sample performance: 5 Jan 2021-29 Dec 2023 | Latest target holdings: 4 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
15.2%	21.4%	0.768	-27.6%	\$1.526



Method: Sentiment Tilt | Estimation: 252 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested; crypto targets unchanged; equity sleeve conserved | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning | Tilt strength: 0.50 | Missing signal: multiplier one; retain missing value

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

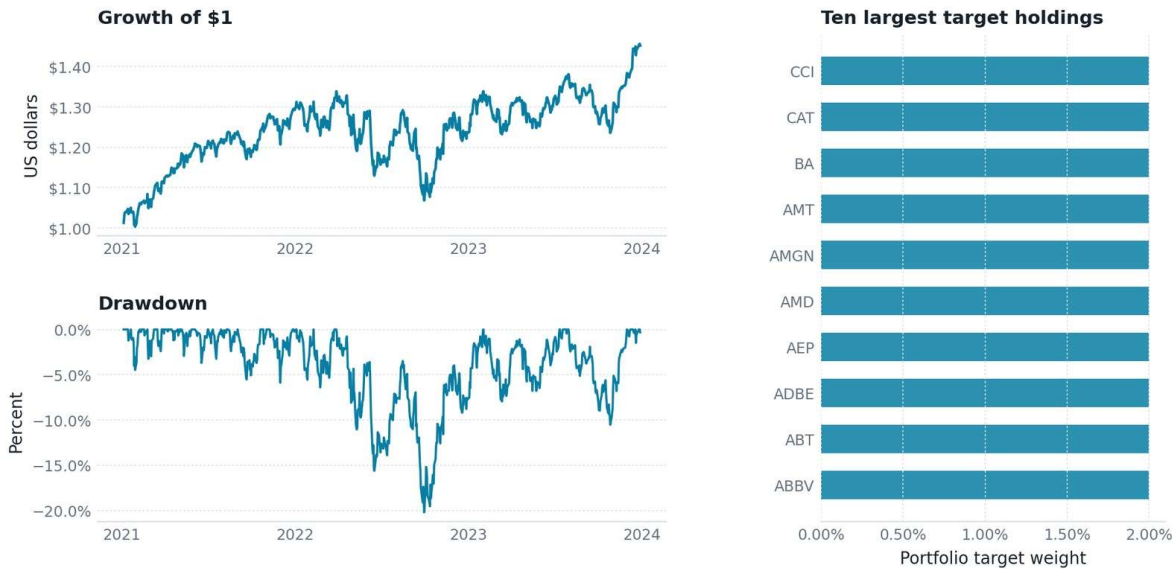
Appendix Figure A4. Sentiment-Tilted Equal Weight fact sheet, 5 Jan 2021-29 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Equity-Only Equal Weight

Out-of-sample performance: 5 Jan 2021-29 Dec 2023 | Latest target holdings: 4 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
13.3%	16.1%	0.855	-20.2%	\$1.451



Method: Equal Weight | Estimation: 252 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

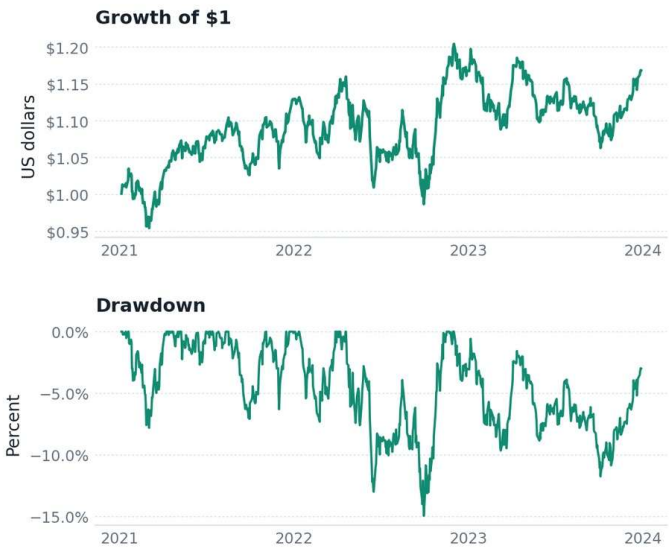
Appendix Figure A5. Equity-Only Equal Weight fact sheet, 5 Jan 2021-29 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Equity-Only Minimum Variance

Out-of-sample performance: 5 Jan 2021-29 Dec 2023 | Latest target holdings: 4 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
5.4%	12.7%	0.473	-14.9%	\$1.168



Method: Min Variance | Estimation: 252 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

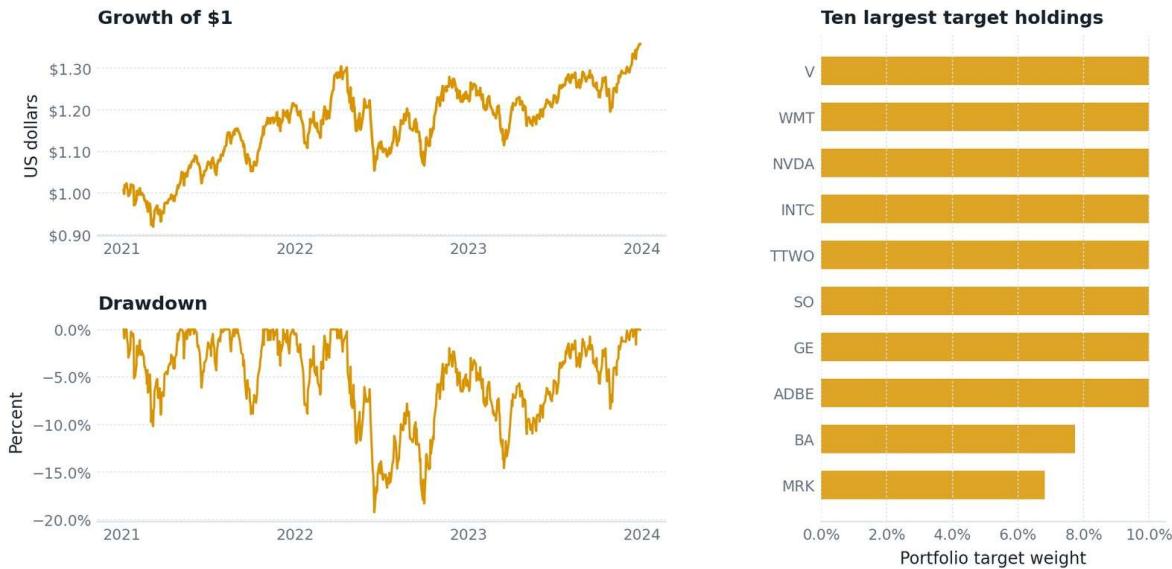
Appendix Figure A6. Equity-Only Minimum Variance fact sheet, 5 Jan 2021-29 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Equity-Only Maximum Sharpe

Out-of-sample performance: 5 Jan 2021-29 Dec 2023 | Latest target holdings: 4 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
10.8%	16.6%	0.700	-19.2%	\$1.359



Method: Max Sharpe | Estimation: 252 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested; maximum 10% per asset | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

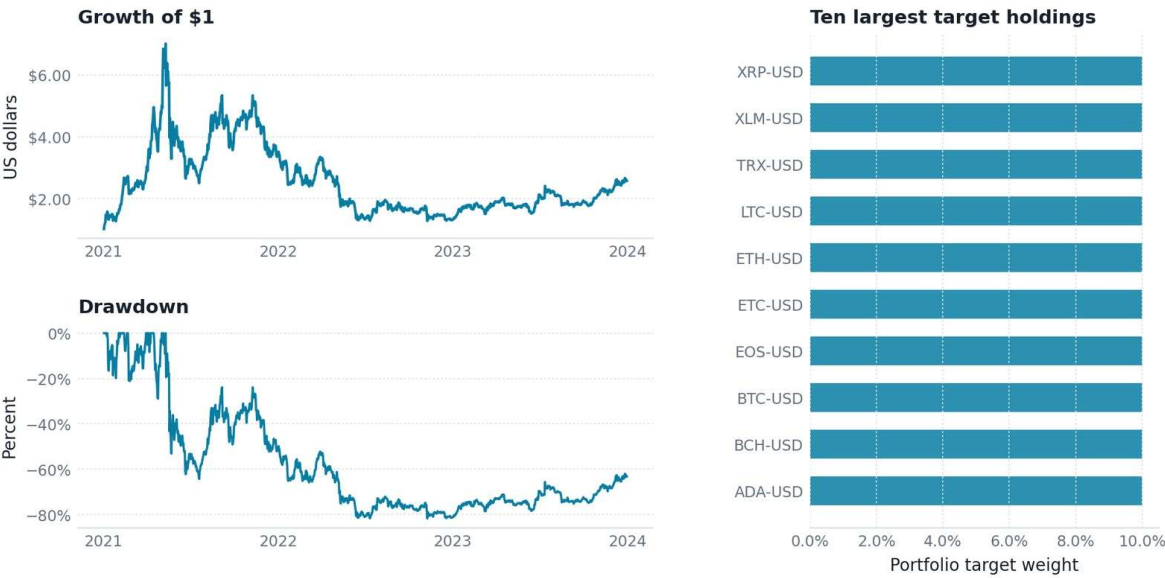
Appendix Figure A7. Equity-Only Maximum Sharpe fact sheet, 5 Jan 2021-29 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Crypto-Only Equal Weight

Out-of-sample performance: 2 Jan 2021-31 Dec 2023 | Latest target holdings: 2 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
37.1%	81.7%	0.799	-81.7%	\$2.576



Method: Equal Weight | Estimation: 365 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

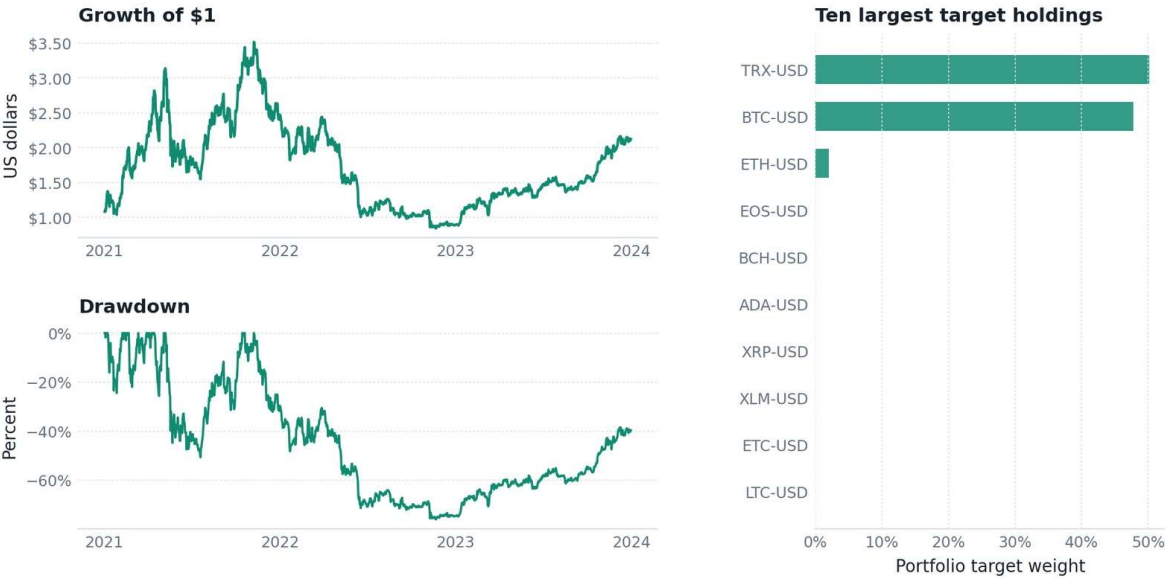
Appendix Figure A8. Crypto-Only Equal Weight fact sheet, 2 Jan 2021-31 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Crypto-Only Minimum Variance

Out-of-sample performance: 2 Jan 2021-31 Dec 2023 | Latest target holdings: 2 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
28.5%	64.7%	0.713	-76.0%	\$2.122



Method: Min Variance | Estimation: 365 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

Appendix Figure A9. Crypto-Only Minimum Variance fact sheet, 2 Jan 2021-31 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / FUND FACT SHEET

Crypto-Only Maximum Sharpe

Out-of-sample performance: 2 Jan 2021-31 Dec 2023 | Latest target holdings: 2 Dec 2023

ANNUAL RETURN	VOLATILITY	SHARPE	MAX DRAWDOWN	GROWTH OF \$1
37.1%	81.7%	0.799	-81.7%	\$2.576



Method: Max Sharpe | Estimation: 365 trading days | Rebalance: monthly, first trading day; effective next trading day

Constraints: long-only; fully invested; maximum 10% per asset | Risk-free rate: 0% | Transaction costs: 0%

Parameters: fixed ex ante; no out-of-sample tuning

Source: results/data/fund_returns.csv, results/data/fund_weights.csv and results/tables/performance_metrics.csv; author calculations. Not investment advice.

Appendix Figure A10. Crypto-Only Maximum Sharpe fact sheet, 2 Jan 2021-31 Dec 2023. Source: results/tables/fact_sheet_summary.csv and latest-holdings CSVs; author calculations.

FINMOSAIC / DRAWDOWN COMPARISON

Minimum Variance limited the deepest loss

Percentage decline from each fund's previous growth-of-\$1 peak



Source: Hosted course price and headline data; author calculations. Out-of-sample period: 5 Jan 2021-29 Dec 2023. Units: percent

Appendix Figure A11. Drawdown comparison for the four combined funds, 5 Jan 2021-29 Dec 2023. Source: results/data/fund_returns.csv; author calculations.

FINMOSAIC / RISK-ADJUSTED PERFORMANCE

Maximum Sharpe led the fund comparison

Annualized mean daily return divided by volatility; risk-free rate assumed zero



Source: Hosted course price and headline data; author calculations. Out-of-sample period: 5 Jan 2021-29 Dec 2023. Units: ratio

Appendix Figure A12. Sharpe-ratio comparison for the four combined funds, 5 Jan 2021-29 Dec 2023. Source: results/tables/performance_metrics.csv; author calculations.

FINMOSAIC / SENTIMENT COVERAGE

News availability differed across equity sectors

Share of equity trading days with at least one covered ticker; no-news days are not neutral



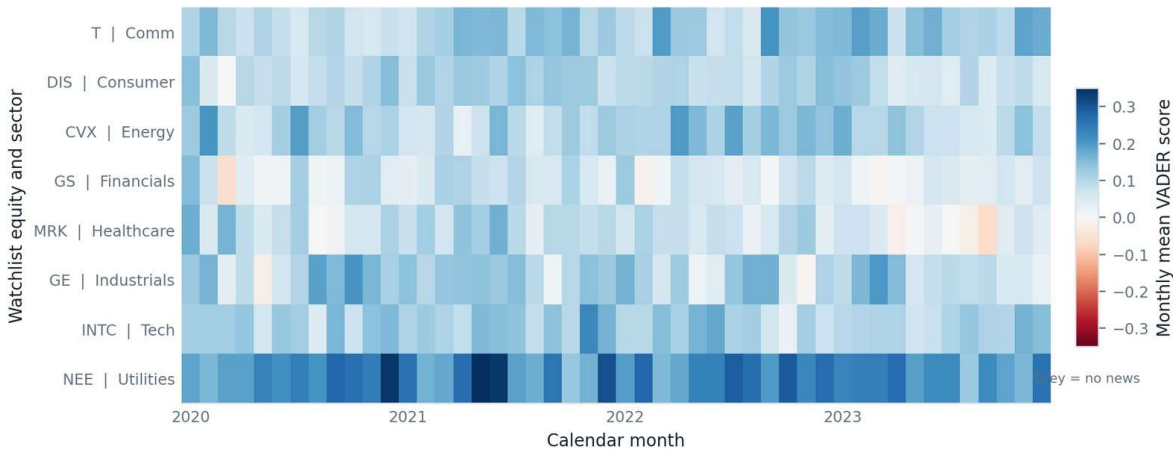
Source: Hosted course headline data; author calculations. Sample: 2 Jan 2020-29 Dec 2023. Coverage is measured before the trading-day lag.

Appendix Figure A13. Observed and missing VADER sector-days, 2 Jan 2020-29 Dec 2023. Source: results/tables/sentiment_summary_by_sector.csv; author calculations.

FINMOSAIC / MONTHLY COMPANY SENTIMENT WATCHLIST

Monthly news tone across eight representative equities

Watchlist: T, DIS, CVX, GS, MRK, GE, INTC, NEE. Highest-coverage ticker from each of the eight largest sectors by headline count.



Source: results/data/company_sentiment_index.csv; hosted course headlines; NLTK VADER; author calculations.
Sample: Jan 2020-Dec 2023. Monthly means use observed ticker-days only; no-news months remain missing.

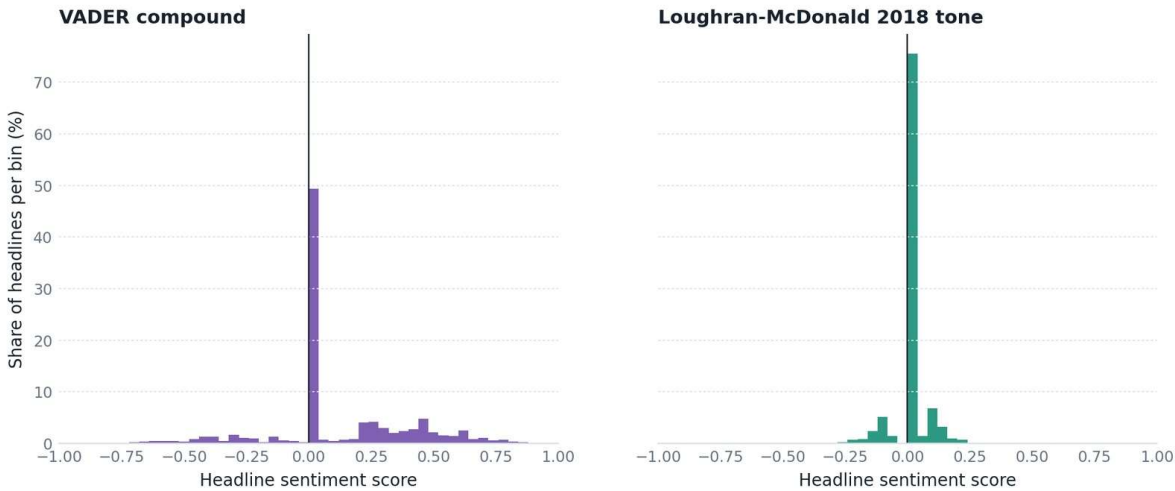
Units: mean VADER score

Appendix Figure A14. Monthly observed-only company VADER sentiment for the stated cross-sector watchlist, Jan 2020-Dec 2023.
Source: results/data/company_sentiment_index.csv; author calculations.

FINMOSAIC / FINANCIAL-DICTIONARY ROBUSTNESS

The finance dictionary changed headline-score shape

Original headline text; fixed model-native scoring; no portfolio use or OOS tuning



Source: Hosted course headlines; NLTK VADER; official Loughran-McDonald 2018 word lists; author calculations.
Sample: headline dates aligned over 2 Jan 2020-29 Dec 2023.

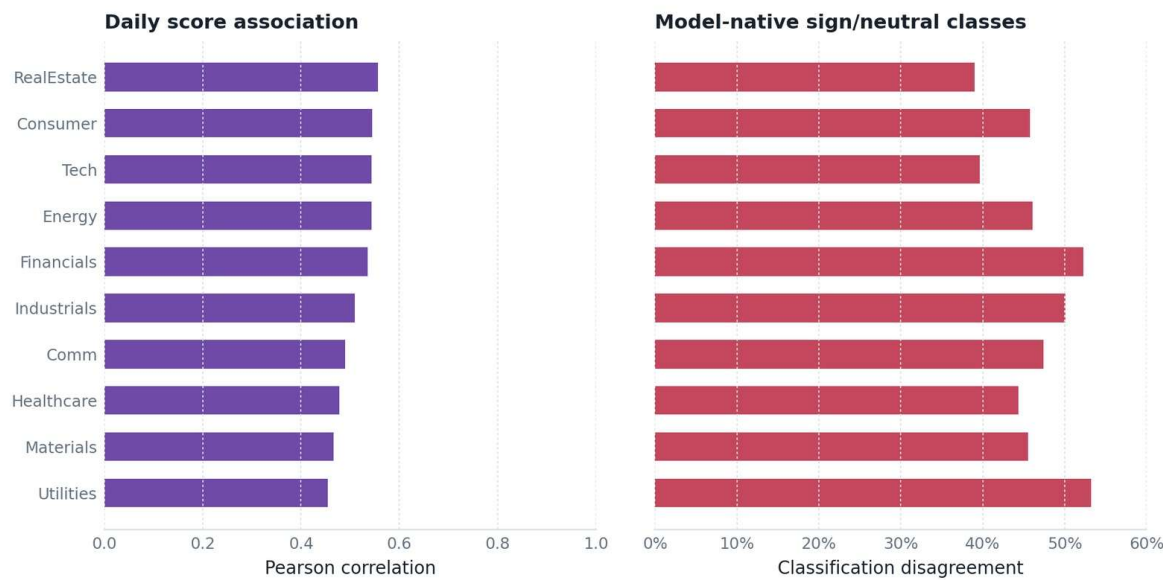
Units: percent; scores [-1, 1]

Appendix Figure A15. Headline-score distributions for VADER and Loughran-McDonald, 2020-2023. Source: results/tables/sentiment_score_distribution_comparison.csv; author calculations.

FINMOSAIC / SECTOR MODEL COMPARISON

VADER and financial tone were related but not interchangeable

Same observed sector-days; equal-weight ticker aggregation; contemporaneous scores



Source: results/data/sector_sentiment_index.csv and results/data/lm_sector_sentiment_index.csv; author calculations.
Sample: 2 Jan 2020-29 Dec 2023. VADER thresholds: +/-0.05; LM threshold: zero.

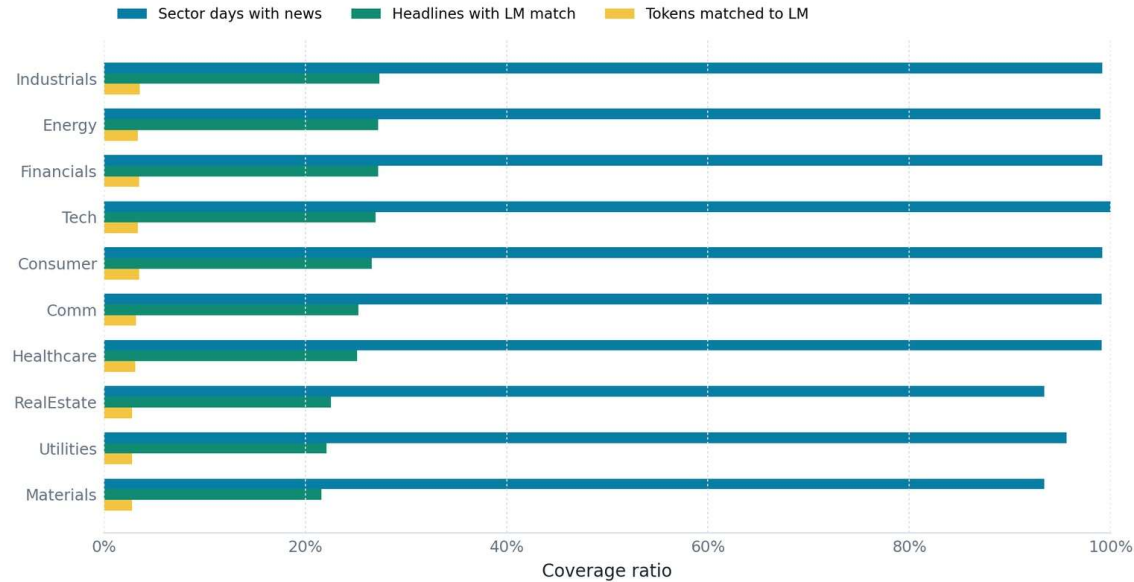
Units: ratio

Appendix Figure A16. Sector correlation and classification disagreement between VADER and Loughran-McDonald, 2020-2023.
Source: results/tables/sentiment_model_sector_comparison.csv; author calculations.

FINMOSAIC / FINANCIAL-LEXICON COVERAGE

Most news days were observed, but sentiment words were sparse

Coverage is disclosed separately from tone; unmatched headlines remain observed zero tone



Source: Hosted course headlines and official Loughran-McDonald 2018 sentiment word lists; author calculations. Sample: 2 Jan 2020-29 Dec 2023. Units: percent

Appendix Figure A17. Loughran-McDonald matched-headline and matched-token coverage by sector, 2020-2023. Source: results/tables/lm_coverage_by_sector.csv; author calculations.